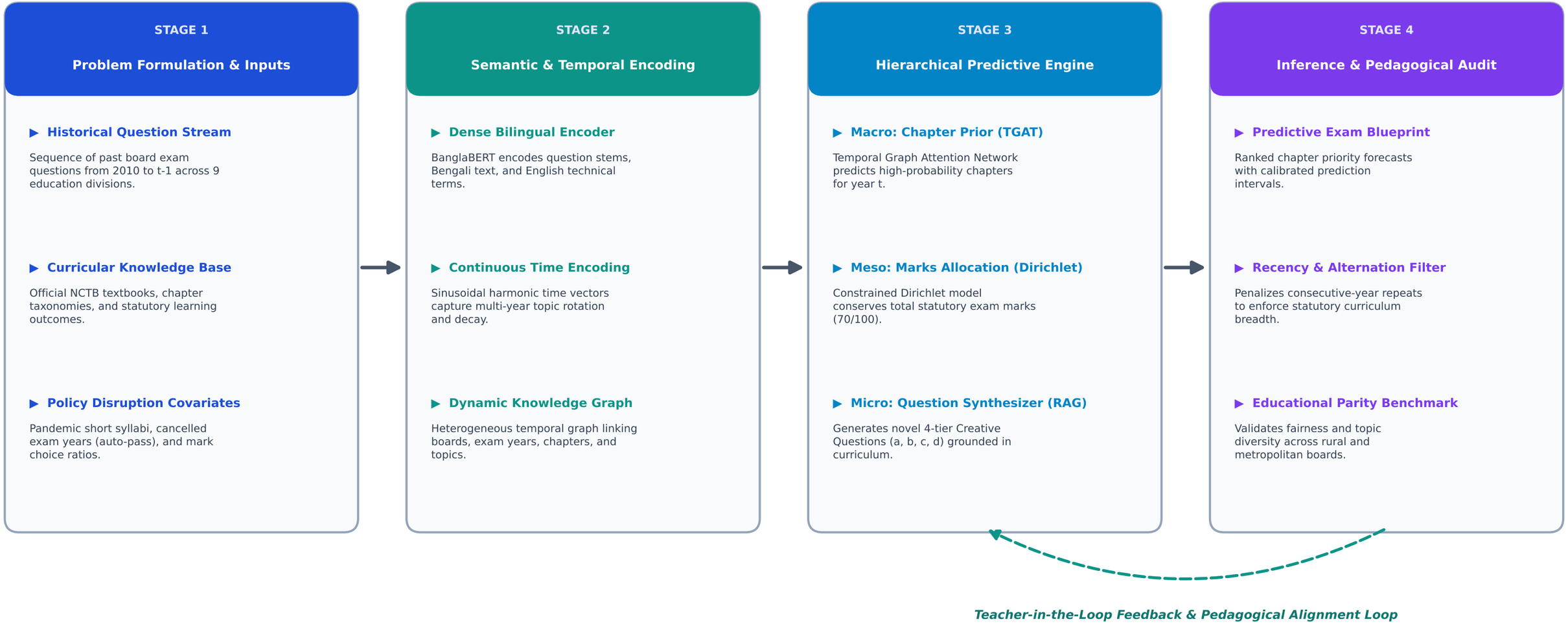
STAGE 1
Problem Formulation & Inputs
▶ Historical Question Stream
Sequence of past board exam questions from 2010 to t-1 across 9 education divisions.
▶ Curricular Knowledge Base
Official NCTB textbooks, chapter taxonomies, and statutory learning outcomes.
▶ Policy Disruption Covariates
Pandemic short syllabi, cancelled exam years (auto-pass), and mark choice ratios.
STAGE 2
Semantic & Temporal Encoding
▶ Dense Bilingual Encoder
BanglaBERT encodes question stems, Bengali text, and English technical terms.
▶ Continuous Time Encoding
Sinusoidal harmonic time vectors capture multi-year topic rotation and decay.
▶ Dynamic Knowledge Graph
Heterogeneous temporal graph linking boards, exam years, chapters, and topics.
STAGE 3
Hierarchical Predictive Engine
▶ Macro: Chapter Prior (TGAT)
Temporal Graph Attention Network predicts high-probability chapters for year t.
▶ Meso: Marks Allocation (Dirichlet)
Constrained Dirichlet model conserves total statutory exam marks (70/100).
▶ Micro: Question Synthesizer (RAG)
Generates novel 4-tier Creative Questions (a, b, c, d) grounded in curriculum.
STAGE 4
Inference & Pedagogical Audit
▶ Predictive Exam Blueprint
Ranked chapter priority forecasts with calibrated prediction intervals.
▶ Recency & Alternation Filter
Penalizes consecutive-year repeats to enforce statutory curriculum breadth.
▶ Educational Parity Benchmark
Validates fairness and topic diversity across rural and metropolitan boards.
Teacher-in-the-Loop Feedback & Pedagogical Alignment Loop